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**Persistence Landscapes
for Heterogeneous Treatment Effects:
Stability and Consistency under
Piecewise-Linear Approximation**

Bachelor's Thesis

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Abstract

This thesis develops a framework connecting causal inference with topological data analysis. The conditional average treatment effect (CATE) function $\tau_0 : \Omega \rightarrow \mathbb{R}$ on a compact domain $\Omega \subset \mathbb{R}^d$ is studied as a topological object: persistent homology is applied to its superlevel-set filtration, and the resulting information is summarized by persistence landscapes.

The main contribution is a quantitative stability and consistency result. If $\tau_0 \in C^{0,\alpha}(\Omega)$ is a tame Hölder function and $\hat{\tau}_n$ is a uniformly consistent estimator with rate r_n , then the persistence landscapes of the piecewise-linear interpolation $\hat{\tau}_{n,h}$ on a shape-regular triangulation with mesh size h satisfy

$$\|\Lambda_k(-\hat{\tau}_{n,h}) - \Lambda_k(-\tau_0)\|_{L^p(\mathbb{R})} = O_P(r_n + h^\alpha).$$

This bound is obtained by composing four individually known results — a Hölder interpolation estimate, the bottleneck stability theorem for persistence diagrams, the L^∞ -stability of persistence landscapes, and an elementary support estimate — into a single end-to-end guarantee for the causal-topological pipeline.

The framework is validated empirically on five synthetic scenarios with known topological structure in dimension $d = 2$. The primary focus is on homological degree $q = 0$ (connected components of regions with elevated treatment effect) and low-dimensional covariate spaces ($d \leq 4$).

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1 Introduction

1.1 Motivation

Causal inference studies the effect of interventions on outcomes. The standard framework is based on potential outcomes [1]; a comprehensive modern treatment is given by Hernán and Robins [2]. In many applications, average treatment effects are insufficient: the effect of an intervention varies across individuals depending on their covariates. This motivates the study of the conditional average treatment effect (CATE)

$$\tau_0(x) = E[Y(1) - Y(0) \mid X = x], \quad x \in \Omega.$$

Flexible nonparametric estimators of τ_0 are now available [3, 4], but they typically treat the estimated function pointwise. As a result, they provide limited information about its *global* structure: whether regions of large treatment effect form a single connected cluster or several separated ones, and whether such structure is stable under estimation error.

1.2 Topological approach

Persistent homology [5, 8] tracks the appearance and disappearance of topological features across a range of thresholds and satisfies strong stability properties [6]. Persistence landscapes, introduced by Bubenik [7], provide a functional representation of persistent homology in a Banach space, enabling both statistical inference and quantitative comparison.

The idea of this thesis is to apply these tools to the superlevel-set filtration of τ_0 . For each threshold $t \in \mathbb{R}$, the superlevel set $\{x \in \Omega : \tau_0(x) \geq t\}$ describes the region where the treatment effect exceeds t . As t varies, the topology of these regions changes: connected components appear, merge, and disappear. Persistence landscapes record this evolution quantitatively.

From a practical standpoint, this perspective enables the identification of *optimal treatment zones*. In personalized medicine or targeted advertising, it is important to determine whether individuals with high expected treatment effects form a single cohesive subgroup or several distinct clusters. Persistent homology provides a quantitative measure of the robustness of such groups: features that persist across a wide range of thresholds represent genuine structure rather than estimation artifacts.

1.3 Contribution and structure

The mathematical content of the thesis connects three layers:

1. **Statistical:** the CATE is estimated from data via cross-fitted residualization (Section 2).
2. **Geometric:** the estimate is interpolated on a shape-regular triangulation of the covariate space (Section 3).
3. **Topological:** persistence landscapes capture the global organization of treatment heterogeneity (Section 3).

The main result (Theorem 5.12) provides a quantitative bound on the landscape error in terms of the statistical estimation rate and the mesh size; the precise statement and proof are given in Section 5. Empirical validation on synthetic data is presented in Section 6.

1.4 Scope

The geometric-topological analysis is intended for low-dimensional covariate spaces ($d \leq 4$), where triangulation is computationally feasible and the interpretation of connected regions is meaningful. The primary focus is on homological degree $q = 0$, although the formal framework applies to all degrees.

2 Causal Framework

2.1 Setup and potential outcomes

Let $W_i = (Y_i, A_i, X_i)$, $i = 1, \dots, n$, be i.i.d. observations, where $Y_i \in \mathbb{R}$ is the observed outcome, $A_i \in \{0, 1\}$ is the binary treatment indicator, and $X_i \in \Omega$ is the vector of covariates. The conditional average treatment effect is defined by

$$\tau_0(x) = E[Y(1) - Y(0) \mid X = x], \quad x \in \Omega,$$

where $Y(0), Y(1)$ denote potential outcomes under the Rubin causal model [1].

2.2 Partially linear model

We work within the partially linear model

$$Y = g_0(X) + A\tau_0(X) + U, \quad E[U \mid X, A] = 0, \quad (1)$$

where g_0 is the baseline response and U is an unobserved error. The causal interpretation of τ_0 requires the standard identification assumptions: consistency ($Y = AY(1) + (1 - A)Y(0)$), conditional ignorability ($(Y(1), Y(0)) \perp\!\!\!\perp A \mid X$), and overlap ($0 < P(A = 1 \mid X = x) < 1$). These are stated here for completeness and are assumed throughout; they are not directly used in the topological analysis that follows.

2.3 Estimation via cross-fitted residualization

Define the nuisance functions $m_0(x) = E[Y \mid X = x]$ and $e_0(x) = P(A = 1 \mid X = x)$. Using K -fold cross-fitting, we obtain out-of-fold estimates $\hat{m}^{(-k(i))}$ and $\hat{e}^{(-k(i))}$ and form residualized variables

$$\tilde{Y}_i = Y_i - \hat{m}^{(-k(i))}(X_i), \quad \tilde{A}_i = A_i - \hat{e}^{(-k(i))}(X_i).$$

The second-stage estimator $\hat{\tau}_n$ is then obtained by minimizing a penalized weighted least-squares objective:

$$\hat{\tau}_n \in \arg \min_{\tau \in \mathcal{F}_n} \left[\frac{1}{n} \sum_{i=1}^n (\tilde{Y}_i - \tilde{A}_i \tau(X_i))^2 + \text{pen}_n(\tau) \right].$$

The specific form of $\mathcal{F}_n$ and pen_n depends on the chosen method. For the purposes of this thesis, all properties of $\hat{\tau}_n$ are encapsulated in a single assumption:

Assumption 1 (Uniform consistency). There exists a sequence $r_n \rightarrow 0$ such that

$$\|\hat{\tau}_n - \tau_0\|_{L^\infty(\Omega)} = O_P(r_n)$$

This is the only statistical assumption used in the main results. It is known to hold for kernel estimators and spline/series estimators under standard conditions, for ensemble methods such as gradient boosting, it is empirically observed but not proven in general (see Section 7 for further remarks).

3 Geometric and Topological Setting

3.1 Covariate space and triangulation

Definition 3.1 (Covariate space). The covariate space is $\mathcal{X} = \Omega \subset \mathbb{R}^d$, where Ω is a bounded domain. Throughout the thesis, we assume Ω is a compact polyhedral domain (its boundary consists of finitely many flat faces), so that it can be exactly covered by a simplicial triangulation without boundary approximation error.

Definition 3.2 (Shape-regular triangulation). A family $\{\mathcal{T}_h\}_{h>0}$ of simplicial triangulations of Ω is called *shape-regular* if there exists a constant $\sigma > 0$ such that for every simplex $T \in \mathcal{T}_h$:

$$\frac{h_T}{\rho_T} \leq \sigma,$$

where $h_T = \text{diam}(T)$ is the diameter and ρ_T is the inradius of T . The mesh size is $h = \max_{T \in \mathcal{T}_h} h_T$.

Shape-regularity means that simplices do not degenerate (become infinitely “thin”) as $h \rightarrow 0$. It guarantees the existence of triangulation families with $h \rightarrow 0$ and is standard in finite element theory.

3.2 Hölder spaces

Definition 3.3 (Hölder space $C^{0,\alpha}(\Omega)$). Let $\Omega \subset \mathbb{R}^d$ be compact and $\alpha \in (0, 1]$. The space $C^{0,\alpha}(\Omega)$ consists of continuous functions $f : \Omega \rightarrow \mathbb{R}$ for which the *Hölder seminorm* is finite:

$$[f]_{C^{0,\alpha}(\Omega)} = \sup_{\substack{x,y \in \Omega \\ x \neq y}} \frac{|f(x) - f(y)|}{\|x - y\|^\alpha}.$$

The norm is $\|f\|_{C^{0,\alpha}} = \|f\|_{L^\infty(\Omega)} + [f]_{C^{0,\alpha}(\Omega)}$, where $\|f\|_{L^\infty(\Omega)} = \sup_{x \in \Omega} |f(x)|$. With this norm, $C^{0,\alpha}(\Omega)$ is a Banach space.

Remark 3.4. The condition $\tau_0 \in C^{0,\alpha}(\Omega)$ means that the treatment effect depends continuously on covariates: individuals with similar characteristics have similar effects. This is a standard and mild assumption in nonparametric statistics, satisfied in most applications where the causal mechanism is governed by continuous physical, biological, or economic processes.

3.3 Barycentric coordinates and piecewise-linear interpolation

Definition 3.5 (Piecewise-linear interpolation). Let $f : \Omega \rightarrow \mathbb{R}$ be a continuous function and $\mathcal{T}_h$ a triangulation of Ω . The *piecewise-linear interpolation* $f_h : \Omega \rightarrow \mathbb{R}$ is the unique continuous function that

- (i) agrees with f at every vertex of $\mathcal{T}_h$, and
- (ii) is affine on each simplex $T \in \mathcal{T}_h$.

Explicitly, on a simplex T with vertices $v_0, \dots, v_d$,

$$f_h(x) = \sum_{i=0}^d \lambda_i(x) f(v_i),$$

where $\lambda_0(x), \dots, \lambda_d(x)$ are the standard barycentric coordinates of x in T (satisfying $\lambda_i \geq 0, \sum_i \lambda_i = 1$).

3.4 Superlevel sets and filtration convention

The topology of the treatment effect function is studied through its superlevel sets.

Definition 3.6 (Superlevel set). For a function $f : \mathcal{X} \rightarrow \mathbb{R}$ and $t \in \mathbb{R}$:

$$A_t(f) = \{x \in \mathcal{X} : f(x) \geq t\}.$$

The family $\{A_t(f)\}_{t \in \mathbb{R}}$ is decreasing in t .

Definition 3.7 (Sublevel set). For a function $f : \mathcal{X} \rightarrow \mathbb{R}$ and $s \in \mathbb{R}$:

$$B_s(f) = \{x \in \mathcal{X} : -f(x) \leq s\}.$$

Under the change of variable $s = -t$, the filtrations $\{A_t(f)\}_{t \in \mathbb{R}}$ and $\{B_s(f)\}_{s \in \mathbb{R}}$ are equivalent.

Throughout the thesis, persistent homology and persistence landscapes associated with superlevel sets of f are understood via the sublevel filtration of $-f$.

3.5 Persistent homology

Definition 3.8 (Betti numbers). For a topological space Z , let $H_q(Z)$ denote its q -th homology group (with coefficients in a fixed field $\mathbb{F}$). Its rank $\beta_q(Z) = \text{rank } H_q(Z)$ is called the q -th *Betti number*. In particular, $\beta_0(Z)$ counts connected components and $\beta_1(Z)$ counts one-dimensional holes.

Definition 3.9 (Tame function). A function $f : X \rightarrow \mathbb{R}$ on a compact topological space is called *tame* if:

- (i) for every $t \in \mathbb{R}$, the homology groups $H_q(\{f \leq t\}; \mathbb{F})$ are finitely generated for all $q \geq 0$;
- (ii) there are only finitely many *critical values*, i.e., values of t at which the homological type of $\{f \leq t\}$ changes.

Definition 3.10 (Persistence diagram). Let $f : X \rightarrow \mathbb{R}$ be a tame function on a compact topological space. The sublevel-set filtration $\{X_t^f\}_{t \in \mathbb{R}}$ gives rise to a family of homology groups with inclusion-induced maps. The *persistence diagram* $\text{Dgm}_q(f)$ (in degree q) is the finite multiset of points $(b_i, d_i) \in \mathbb{R}^2$, $b_i < d_i < +\infty$, where b_i is the birth value and d_i is the death value of a homological class. All diagonal points $\{(t, t) : t \in \mathbb{R}\}$ are included with infinite multiplicity.

Convention. Homological classes that never die (i.e., persist to $t = +\infty$) are called *essential classes* and are excluded from $\text{Dgm}_q(f)$. In degree $q = 0$ on a connected compact space, there is exactly one such class (the global connected component). Since essential classes are common to both f and any sufficiently close perturbation, their exclusion does not affect the stability results below.

The quantity $d_i - b_i > 0$ is called the *persistence* (or lifetime) of the feature (b_i, d_i) . Features with large persistence represent robust topological structures; features with small persistence are typically noise.

Definition 3.11 (Bottleneck distance). For two persistence diagrams D_1, D_2 (including diagonal points):

$$d_\infty(D_1, D_2) = \inf_{\gamma: D_1 \rightarrow D_2} \sup_{p \in D_1} \|p - \gamma(p)\|_\infty,$$

where the infimum is over all bijections γ and $\|(a, b) - (c, d)\|_\infty = \max(|a - c|, |b - d|)$.

3.6 Persistence landscapes

Definition 3.12 (Tent function). Given a birth-death pair (b, d) with $b < d$, define

$$f_{(b,d)}(t) = \max(0, \min(t - b, d - t)), \quad t \in \mathbb{R}.$$

This is a piecewise-linear function that is zero outside (b, d) , rises linearly from 0 to $\frac{d-b}{2}$ on $[b, \frac{b+d}{2}]$, and descends linearly back to 0 on $[\frac{b+d}{2}, d]$.

Definition 3.13 (Persistence landscape (Bubenik, 2015)). Let $D = \{(b_i, d_i)\}_{i=1}^N$ be a finite persistence diagram (without diagonal points). For each $t \in \mathbb{R}$, consider the values $\{f_{(b_i, d_i)}(t)\}_{i=1}^N$ and order them in decreasing order. The k -th largest value is denoted $\lambda_k^f(t)$, and the sequence $\Lambda_q(f) = \{\lambda_k^f\}_{k \geq 1}$ is called the *persistence landscape* of f in degree q . If fewer than k values are positive, we set $\lambda_k^f(t) = 0$.

Remark 3.14. Since essential classes (with $d_i = +\infty$) are excluded from the persistence diagram (Definition 3.10), the diagram is a finite multiset of points in $\mathbb{R}^2$. Each tent function $f_{(b_i, d_i)}$ therefore has compact support, and consequently each landscape function λ_k^f is continuous, piecewise-linear, and compactly supported. In particular, $\|\lambda_k^f\|_{L^p(\mathbb{R})} < \infty$ for all $1 \leq p \leq \infty$.

Definition 3.15 (L^p -norm of persistence landscapes). For $1 \leq p < \infty$:

$$\|\Lambda_q(f) - \Lambda_q(g)\|_p = \left(\sum_{k=1}^{\infty} \int_{\mathbb{R}} |\lambda_k^f(t) - \lambda_k^g(t)|^p dt \right)^{1/p}.$$

For $p = \infty$:

$$\|\Lambda_q(f) - \Lambda_q(g)\|_\infty = \sup_{k \geq 1} \sup_{t \in \mathbb{R}} |\lambda_k^f(t) - \lambda_k^g(t)|.$$

For a fixed layer k : $\|\lambda_k^f - \lambda_k^g\|_{L^p(\mathbb{R})} = \left(\int_{\mathbb{R}} |\lambda_k^f(t) - \lambda_k^g(t)|^p dt \right)^{1/p}$.

Remark 3.16. For landscapes built from finite diagrams of tame functions on compact spaces, each λ_k^f is a continuous piecewise-linear function with compact support. Hence both L^p -norms are finite.

4 Assumptions

4.1 Geometric assumptions

Assumption 2 (Domain). $\Omega \subset \mathbb{R}^d$ is a compact polyhedral domain.

Assumption 3 (Triangulation). $\{\mathcal{T}_h\}_{h>0}$ is a shape-regular family of simplicial triangulations of Ω with mesh size $h = \max_T \text{diam}(T)$.

4.2 Regularity assumptions on τ_0

Assumption 4 (Hölder regularity). $\tau_0 \in C^{0,\alpha}(\Omega)$ for some $\alpha \in (0, 1]$ with Hölder semi-norm $L = [\tau_0]_{C^{0,\alpha}} < \infty$.

Assumption 5 (Tameness). The function $-\tau_0 : \Omega \rightarrow \mathbb{R}$ is tame (Definition 3.9).

4.3 Statistical assumptions

Assumption 6 (Fixed parameters). Fixed: $p \in [1, +\infty]$, homological degree $q \geq 0$, landscape layer $k \geq 1$.

Here and throughout, $Z_n = O_P(r_n)$ means that Z_n/r_n is bounded in probability, i.e., for every $\varepsilon > 0$ there exist M, N such that $P(|Z_n| > M r_n) < \varepsilon$ for all $n \geq N$.

Assumption 7 (Uniform consistency of the estimator). There exists a sequence $r_n \rightarrow 0$ such that

$$\|\hat{\tau}_n - \tau_0\|_{L^\infty(\Omega)} = O_P(r_n).$$

Remark 4.1. Uniform convergence in L^∞ -norm is a strong requirement. It is established for kernel and series estimators under standard smoothness conditions. For gradient boosting and random forests, analogous results in sup-norm are not available in general, although the behavior is observed empirically.

5 Main Results

The proof is organized into three parts:

1. **Part I (Geometric estimate):** bounding the error from replacing τ_0 by its piecewise-linear interpolation $\tau_{0,h}$.
2. **Part II (Statistical estimate):** bounding the error from replacing τ_0 by its statistical estimate $\hat{\tau}_n$.
3. **Part III (Assembly):** combining both estimates via the triangle inequality to obtain the main theorem.

The overall scheme:

$$\underbrace{\|\tau_{0,h} - \tau_0\|_\infty \leq L h^\alpha}_{\text{Lemma 5.1}} \Rightarrow \underbrace{d_\infty \leq L h^\alpha}_{\text{Lemma 5.2}} \Rightarrow \underbrace{\|\Lambda_k\|_\infty \leq L h^\alpha}_{\text{Lemma 5.3}} \Rightarrow \underbrace{\|\Lambda_k\|_p \leq C h^\alpha}_{\text{Lemma 5.4}} \\ \xrightarrow{+ \text{Lemma 5.6} + \text{triangle inequality}} \|\Lambda_k(-\hat{\tau}_{n,h}) - \Lambda_k(-\tau_0)\|_p = O_P(r_n + h^\alpha).$$

5.1 Part I: Geometric Estimate

Lemma 5.1 (Interpolation estimate). *Let Assumptions 2, 3, 4 hold. Then*

$$\|\tau_{0,h} - \tau_0\|_{L^\infty(\Omega)} \leq L h^\alpha, \quad (2)$$

where $L = [\tau_0]_{C^{0,\alpha}}$ and $h = \max_T \text{diam}(T)$.

Proof. Fix an arbitrary simplex $T \in \mathcal{T}_h$ with vertices $v_0, \dots, v_d$ and an arbitrary point $x \in T$.

By Definition 3.5:

$$\tau_{0,h}(x) = \sum_{i=0}^d \lambda_i(x) \tau_0(v_i).$$

Using the partition of unity $\sum_i \lambda_i(x) = 1$:

$$\tau_{0,h}(x) - \tau_0(x) = \sum_{i=0}^d \lambda_i(x) \tau_0(v_i) - \tau_0(x) \sum_{i=0}^d \lambda_i(x) = \sum_{i=0}^d \lambda_i(x) [\tau_0(v_i) - \tau_0(x)].$$

Using triangle inequality

$$|\tau_{0,h}(x) - \tau_0(x)| \leq \sum_{i=0}^d \lambda_i(x) |\tau_0(v_i) - \tau_0(x)|.$$

By Assumption 4 and since $\|v_i - x\| \leq h$:

$$|\tau_0(v_i) - \tau_0(x)| \leq L \|v_i - x\|^\alpha \leq L \cdot h^\alpha$$

Applying this to the inequality above

$$|\tau_{0,h}(x) - \tau_0(x)| \leq \sum_{i=0}^d \lambda_i(x) \cdot L \cdot h^\alpha = L h^\alpha \cdot \underbrace{\sum_{i=0}^d \lambda_i(x)}_{=1} = L h^\alpha.$$

Since $x \in T$ and $T \in \mathcal{T}_h$ are arbitrary: $\|\tau_{0,h} - \tau_0\|_{L^\infty(\Omega)} \leq L h^\alpha$. □

Lemma 5.2 (Stability of persistence diagrams). *Let $f, g : X \rightarrow \mathbb{R}$ be tame functions on a compact topological space X . Then for each homological degree $q \geq 0$:*

$$d_\infty(\text{Dgm}_q(f), \text{Dgm}_q(g)) \leq \|f - g\|_{L^\infty(X)}. \quad (3)$$

Proof. This is the *Stability Theorem* of Cohen-Steiner, Edelsbrunner, and Harer [6].

Set $\varepsilon = \|f - g\|_\infty$. For every $t \in \mathbb{R}$:

$$\{f \leq t - \varepsilon\} \subseteq \{g \leq t\} \subseteq \{f \leq t + \varepsilon\}.$$

Indeed, if $f(x) \leq t - \varepsilon$, then $g(x) \leq f(x) + \varepsilon \leq t$, giving the first inclusion. Similarly for the second.

These inclusions induce, for every $q \geq 0$, an ε -interleaving of the persistence modules $\{H_q(\{f \leq t\})\}_t$ and $\{H_q(\{g \leq t\})\}_t$. By the Algebraic Stability Theorem (see [9]), an ε -interleaving implies $d_\infty(\text{Dgm}_q(f), \text{Dgm}_q(g)) \leq \varepsilon$. $\square$

Application. Set $f = -\tau_0$ (tame by 5), $g = -\tau_{0,h}$ (tame as a PL function on a finite complex). Then:

$$d_\infty(\text{Dgm}_q(-\tau_0), \text{Dgm}_q(-\tau_{0,h})) \leq L h^\alpha. \quad (4)$$

Lemma 5.3 (L^∞ -stability of landscapes). *For finite persistence diagrams D_1, D_2 and each $k \geq 1$:*

$$\|\lambda_k(D_1) - \lambda_k(D_2)\|_{L^\infty(\mathbb{R})} \leq d_\infty(D_1, D_2). \quad (5)$$

Proof. This is Theorem 13 in Bubenik [7].

Let $\varepsilon = d_\infty(D_1, D_2)$. Then there exists a bijection $\gamma : D_1 \rightarrow D_2$ (including diagonal points) such that $\|p - \gamma(p)\|_\infty \leq \varepsilon$ for all $p \in D_1$.

Step 1. Let $p = (b, d) \in D_1$ and $\gamma(p) = (b', d')$ with $|b - b'| \leq \varepsilon$, $|d - d'| \leq \varepsilon$. We claim:

$$|f_{(b,d)}(t) - f_{(b',d')}(t)| \leq \varepsilon \quad \forall t \in \mathbb{R}. \quad (6)$$

To verify, note that $f_{(b,d)}(t) = \max(0, \min(t - b, d - t))$. This is a piecewise-linear function with slopes in $\{-1, 0, +1\}$.

Case 1: If $t \leq b$ and $t \leq b'$, both tent functions are zero.

Case 2: If $b \leq t \leq \frac{b+d}{2}$, then $f_{(b,d)}(t) = t - b$. We have $f_{(b',d')}(t) \leq t - b'$ (if $t \geq b'$) or $f_{(b',d')}(t) = 0$ (if $t < b'$). In either case, $|f_{(b,d)}(t) - f_{(b',d')}(t)| \leq |b - b'| \leq \varepsilon$. The remaining cases are analogous by symmetry.

If p is matched to a diagonal point (c, c) , then $f_{(c,c)}(t) = 0$ and $f_{(b,d)}(t) \leq \frac{d-b}{2} \leq \varepsilon$ (since matching to the diagonal means $\max(|b - c|, |d - c|) \leq \varepsilon$ and $d - b \leq 2\varepsilon$).

Step 2. Fix $t \in \mathbb{R}$. Let $a_1 \geq a_2 \geq \dots$ be the values $\{f_p(t)\}_{p \in D_1}$ in decreasing order, and $b_1 \geq b_2 \geq \dots$ be the values $\{f_{\gamma(p)}(t)\}_{p \in D_1}$ (reindexed by γ , not yet sorted).

By Step 1, $|a_{\pi(i)} - b_i| \leq \varepsilon$ for the identity matching $\pi = \text{id}$. After sorting b -values into decreasing order $b_{(1)} \geq b_{(2)} \geq \dots$, a standard result on order statistics gives:

$$|a_k - b_{(k)}| \leq \varepsilon \quad \forall k.$$

(This follows because sorting is a contraction in ℓ^∞ : if two sequences differ by at most ε entry-wise under some bijection, their sorted versions differ by at most ε .)

Hence $|\lambda_k(D_1)(t) - \lambda_k(D_2)(t)| \leq \varepsilon$ for all t . $\square$

Therefore:

$$\|\Lambda_k(-\tau_{0,h}) - \Lambda_k(-\tau_0)\|_{L^\infty(\mathbb{R})} \leq L h^\alpha. \quad (7)$$

Lemma 5.4 (L^p -bound from L^∞ -bound). *Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function supported in $[a, b]$. Then for $p \in [1, \infty)$:*

$$\|u\|_{L^p(\mathbb{R})} \leq (b-a)^{1/p} \|u\|_{L^\infty(\mathbb{R})}. \quad (8)$$

Proof.

$$\|u\|_{L^p}^p = \int_{-\infty}^{+\infty} |u(t)|^p dt = \int_a^b |u(t)|^p dt \leq (b-a) \|u\|_{L^\infty}^p.$$

Taking the p -th root yields (8). $\square$

Proposition 5.5 (Uniform support bound). *Let $M_0 = \|\tau_0\|_{L^\infty(\Omega)}$. For all h with $L h^\alpha \leq 1$:*

- (i) *The support of $\lambda_k(-\tau_0)$ is contained in $[-M_0, M_0]$.*
- (ii) *The support of $\lambda_k(-\tau_{0,h})$ is contained in $[-(M_0 + 1), M_0 + 1]$.*
- (iii) *The support of their difference is contained in $[-(M_0 + 1), M_0 + 1]$.*

Proof. (i) The values of $-\tau_0$ lie in $[-M_0, M_0]$, so all diagram points (b_i, d_i) satisfy $-M_0 \leq b_i < d_i \leq M_0$. Each tent function $f_{(b_i, d_i)}$ is supported in $(b_i, d_i) \subset [-M_0, M_0]$.

(ii) By (2), $\|\tau_{0,h}\|_\infty \leq \|\tau_0\|_\infty + L h^\alpha \leq M_0 + 1$. The argument is then identical to (i).

(iii) Follows from (i) and (ii). $\square$

Conclusion of Part I. Setting $C_1 = (2(M_0 + 1))^{1/p} \cdot L$:

$$\boxed{\|\lambda_k(-\tau_{0,h}) - \lambda_k(-\tau_0)\|_{L^p(\mathbb{R})} \leq C_1 h^\alpha.} \quad (9)$$

For $p = \infty$, we have $C_1 = L$.

5.2 Part II: Statistical Estimate

Lemma 5.6 (Contractivity of interpolation). *Let $f, g : \Omega \rightarrow \mathbb{R}$ be continuous functions and f_h, g_h their piecewise-linear interpolations on the same triangulation $\mathcal{T}_h$. Then*

$$\|f_h - g_h\|_{L^\infty(\Omega)} \leq \|f - g\|_{L^\infty(\Omega)}. \quad (10)$$

Proof. Fix a simplex T with vertices $v_0, \dots, v_d$ and a point $x \in T$. By Definition 3.5:

$$f_h(x) - g_h(x) = \sum_{i=0}^d \lambda_i(x) f(v_i) - \sum_{i=0}^d \lambda_i(x) g(v_i) = \sum_{i=0}^d \lambda_i(x) [f(v_i) - g(v_i)].$$

Taking absolute values:

$$|f_h(x) - g_h(x)| \leq \sum_{i=0}^d \lambda_i(x) |f(v_i) - g(v_i)| \leq \sum_{i=0}^d \lambda_i(x) \|f - g\|_{L^\infty(\Omega)} = \|f - g\|_{L^\infty(\Omega)}.$$

Since x and T are arbitrary, $\|f_h - g_h\|_{L^\infty(\Omega)} \leq \|f - g\|_{L^\infty(\Omega)}$. $\square$

Remark 5.7. The proof is structurally identical to Lemma 5.1. The key property in both cases is the partition of unity: $\sum_i \lambda_i(x) = 1$, $\lambda_i(x) \geq 0$. In Lemma 5.1, the Hölder condition bounds $|f(v_i) - f(x)|$, here, the sup-norm bounds $|f(v_i) - g(v_i)|$.

Lemma 5.8 (Landscape stability for the statistical error). *Let $\hat{\tau}_n, \tau_0 : \Omega \rightarrow \mathbb{R}$ be continuous functions, and let $\hat{\tau}_{n,h}$ and $\tau_{0,h}$ be their piecewise-linear interpolations on the same triangulation $\mathcal{T}_h$. Then*

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_{0,h})\|_{L^p(\mathbb{R})} \leq C_2 \|\hat{\tau}_n - \tau_0\|_{L^\infty(\Omega)}, \quad (11)$$

where C_2 depends only on $\|\tau_0\|_\infty$ and p , and is bounded for small $\|\hat{\tau}_n - \tau_0\|_\infty$.

Proof. The argument applies Lemmas 5.6–5.4 to the pair $(-\hat{\tau}_{n,h}, -\tau_{0,h})$.

Step 1 (Lemma 5.6):

$$\|\hat{\tau}_{n,h} - \tau_{0,h}\|_{L^\infty(\Omega)} \leq \|\hat{\tau}_n - \tau_0\|_{L^\infty(\Omega)}.$$

Step 2 (Lemma 5.2, applied to the tame functions $-\hat{\tau}_{n,h}$ and $-\tau_{0,h}$):

$$d_\infty(\text{Dgm}_q(-\hat{\tau}_{n,h}), \text{Dgm}_q(-\tau_{0,h})) \leq \|\hat{\tau}_{n,h} - \tau_{0,h}\|_{L^\infty} \leq \|\hat{\tau}_n - \tau_0\|_{L^\infty}.$$

Step 3 (Lemma 5.3):

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_{0,h})\|_{L^\infty} \leq \|\hat{\tau}_n - \tau_0\|_{L^\infty}.$$

Step 4 (Lemma 5.4). The supports of both landscapes are contained in $[-(M_0 + 2), M_0 + 2]$ when $\|\hat{\tau}_n - \tau_0\|_\infty \leq 1$ (analogous to Proposition 5.5, with the additional tolerance for $\hat{\tau}_n$). Hence:

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_{0,h})\|_{L^p} \leq (2(M_0 + 2))^{1/p} \|\hat{\tau}_n - \tau_0\|_{L^\infty}.$$

Set $C_2 = (2(M_0 + 2))^{1/p}$. □

Note. Both $-\hat{\tau}_{n,h}$ and $-\tau_{0,h}$ are piecewise-linear functions on the finite simplicial complex $\mathcal{T}_h$, hence **automatically tame**. Assumption 5 is needed only for the true (non-interpolated) function $-\tau_0$.

5.3 Part III: Assembly

Proposition 5.9 (Error decomposition). *Under Assumptions 2–6 and triangle inequality:*

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p} \leq \underbrace{\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_{0,h})\|_{L^p}}_{\text{statistical error}} + \underbrace{\|\lambda_k(-\tau_{0,h}) - \lambda_k(-\tau_0)\|_{L^p}}_{\text{geometric error}}. \quad (12)$$

Theorem 5.10 (Deterministic error bound). *Let Assumptions 2–6 hold. Set $M_0 = \|\tau_0\|_{L^\infty}$, $L = [\tau_0]_{C^{0,\alpha}}$. Then for all h and $\hat{\tau}_n$ satisfying $L h^\alpha \leq 1$ and $\|\hat{\tau}_n - \tau_0\|_\infty \leq 1$:*

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p(\mathbb{R})} \leq C_2 \|\hat{\tau}_n - \tau_0\|_{L^\infty(\Omega)} + C_1 h^\alpha, \quad (13)$$

where

$$C_1 = (2(M_0 + 1))^{1/p} \cdot L, \quad C_2 = (2(M_0 + 2))^{1/p}.$$

For $p = \infty$: $C_1 = L$, $C_2 = 1$.

Remark 5.11. The conditions $L h^\alpha \leq 1$ and $\|\hat{\tau}_n - \tau_0\|_\infty \leq 1$ are imposed solely for convenience: they simplify the support estimates in Proposition 5.5. Replacing the threshold 1 by any $\delta > 0$ yields the same bound with constants $C_1(\delta)$, $C_2(\delta)$ depending on δ . Since both $h \rightarrow 0$ and $\|\hat{\tau}_n - \tau_0\|_\infty \xrightarrow{P} 0$, these conditions are satisfied for all sufficiently small h and sufficiently large n .

Proof. Substitute into (12):

- the first term is bounded by Lemma 5.8: $\leq C_2 \|\hat{\tau}_n - \tau_0\|_\infty$;
- the second term is bounded by (9): $\leq C_1 h^\alpha$. □

Theorem 5.12 (Main theorem: consistency of persistence landscapes). *Let Assumptions 2–7 hold. Then*

$$\boxed{\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p(\mathbb{R})} = O_P(r_n + h^\alpha).} \quad (14)$$

In particular, if $r_n \rightarrow 0$ and $h \rightarrow 0$, then

$$\lambda_k(-\hat{\tau}_{n,h}) \xrightarrow{L^p, P} \lambda_k(-\tau_0),$$

*i.e., the persistence landscapes of the triangulated estimator converge to the persistence landscapes of the true treatment effect function **in probability** in the L^p -norm.*

Proof. By Theorem 5.10, for sufficiently small $\|\hat{\tau}_n - \tau_0\|_\infty$ and h :

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p} \leq C_2 \|\hat{\tau}_n - \tau_0\|_{L^\infty} + C_1 h^\alpha.$$

By Assumption 7, $\|\hat{\tau}_n - \tau_0\|_{L^\infty} = O_P(r_n)$. Hence $C_2 \|\hat{\tau}_n - \tau_0\|_{L^\infty} = O_P(r_n)$. The second term $C_1 h^\alpha$ is deterministic, hence $C_1 h^\alpha = O_P(h^\alpha)$ trivially (a deterministic quantity is bounded in probability). Therefore:

$$\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p} \leq O_P(r_n) + O_P(h^\alpha) = O_P(r_n + h^\alpha).$$

If $r_n \rightarrow 0$ and $h \rightarrow 0$, then $r_n + h^\alpha \rightarrow 0$, and O_P of a sequence tending to zero tends to zero in probability. □

Remark 5.13 (Convergence notation). The notation $\xrightarrow{L^p, P}$ means convergence in probability with respect to the $L^p(\mathbb{R})$ -norm: for every $\varepsilon > 0$,

$$P(\|\lambda_k(-\hat{\tau}_{n,h}) - \lambda_k(-\tau_0)\|_{L^p} > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty, h \rightarrow 0.$$

6 Empirical Illustrations

The purpose of this section is to empirically validate the theoretical results of Theorem 5.12 on simulated data with known ground-truth topological structure. Since the true treatment effect function τ_0 is specified explicitly in each scenario, we can compute both the true persistence landscapes $\Lambda_k(-\tau_0)$ and the estimated landscapes $\Lambda_k(-\hat{\tau}_{n,h})$, and directly measure the error $\|\Lambda_k(-\hat{\tau}_{n,h}) - \Lambda_k(-\tau_0)\|_{L^p}$ as a function of the sample size n and the mesh size h .

We consider five topological scenarios: one connected component, two components, three components, four components, and an annulus, for each scenario we vary the sample size $n \in \{2000, 4000, 6000, 8000\}$ while keeping the triangulation fixed. This isolates the statistical error and allows us to observe the convergence $O_P(r_n)$ predicted by the theory.

6.1 Data generating process

All five scenarios share the same partially linear model (equation (1)) on the covariate space $\Omega = [0, 1]^2 \subset \mathbb{R}^2$ with $d = 2$.

6.1.1 Common components

Covariates. $X = (X_1, X_2) \sim \text{Uniform}([0, 1]^2)$, i.i.d.

Propensity score. The treatment assignment probability is

$$e_0(x) = \sigma(0.5(x_1 - 0.5) + 0.3(x_2 - 0.5)), \quad (15)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function. This produces mildly heterogeneous propensity scores bounded away from $\{0, 1\}$.

Baseline function.

$$g_0(x) = 0.5 \sin(2\pi x_1) \cos(2\pi x_2). \quad (16)$$

This is a smooth nonlinear function independent of the treatment effect geometry.

Noise. $U \sim \mathcal{N}(0, \sigma_U^2)$ with $\sigma_U = 0.05$, independent of (X, A) .

Observed outcome. $Y = g_0(X) + A\tau_0(X) + U$.

Treatment effect function. The function τ_0 varies across scenarios and is described below. In all cases, τ_0 is a sum of Gaussian bumps of the form

$$b(x; c, s) = \exp\left(-\frac{\|x - c\|^2}{2s^2}\right), \quad (17)$$

where $c \in \mathbb{R}^2$ is the center and $s > 0$ is the bandwidth. Each τ_0 is C^∞ , hence trivially Hölder-regular (for any $\alpha \in (0, 1]$) and tame.

6.1.2 Scenario 1: Single component

$$\tau_0(x) = b(x; (0.5, 0.5), 0.18). \quad (18)$$

A single Gaussian bump centered at $(0.5, 0.5)$. The superlevel sets $\{\tau_0 \geq t\}$ form a single connected component for all $t > 0$.

Expected topology: $\beta_0 = 1$ (one component) for all thresholds. The persistence diagram $\text{Dgm}_0(-\tau_0)$ contains one point with infinite persistence (the global component) and no finite-persistence points.

6.1.3 Scenario 2: Two components

$$\tau_0(x) = b(x; (0.3, 0.35), 0.12) + 0.9 b(x; (0.72, 0.68), 0.14). \quad (19)$$

Two Gaussian bumps of slightly different heights, well-separated in $[0, 1]^2$.

Expected topology: at high thresholds, two isolated components; as t decreases, they eventually merge. The persistence diagram contains one finite-persistence point (corresponding to the shorter bump merging into the taller one).

6.1.4 Scenario 3: Three components

$$\tau_0(x) = b(x; (0.25, 0.3), 0.1) + 0.75 b(x; (0.72, 0.3), 0.1) + 0.5 b(x; (0.5, 0.75), 0.1). \quad (20)$$

Three bumps with strictly decreasing amplitudes 1.00, 0.75, 0.50.

Expected topology: three components at high thresholds, merging sequentially as t decreases. The persistence diagram contains two finite-persistence points with distinct lifetimes.

6.1.5 Scenario 4: Four components

$$\begin{aligned} \tau_0(x) = & b(x; (0.22, 0.22), 0.09) + 0.80 b(x; (0.78, 0.22), 0.09) \\ & + 0.60 b(x; (0.22, 0.78), 0.09) + 0.40 b(x; (0.78, 0.78), 0.09). \end{aligned} \quad (21)$$

Four bumps in the corners of $[0, 1]^2$ with strictly decreasing amplitudes.

Expected topology: four components at high thresholds. Three finite-persistence points in the diagram.

6.1.6 Scenario 5: Annulus

$$\tau_0(x) = \exp\left(-\frac{(\|x - (0.5, 0.5)\| - 0.3)^2}{2 \cdot 0.07^2}\right). \quad (22)$$

A ring-shaped region of elevated treatment effect centered at $(0.5, 0.5)$ with radius ≈ 0.3 and width ≈ 0.07 .

Expected topology: $\beta_0 = 1$ (the ring is connected), $\beta_1 = 1$ (there is a hole inside the ring). This scenario goes beyond the $q = 0$ focus of the thesis and illustrates the potential of higher-degree persistent homology.

Table 1 summarizes all five scenarios.

Table 1: Summary of topological scenarios.

Scenario	Shape	β_0	β_1	Finite points in Dgm_0
Single	one bump	1	0	0
Two blobs	two bumps	2	0	1
Three blobs	three bumps	3	0	2
Four blobs	four bumps	4	0	3
Annulus	ring	1	1	0

6.2 Estimation procedure

6.2.1 CATE estimation

The treatment effect function is estimated by the cross-fitted residualised procedure implemented in three steps.

Step 1: Nuisance estimation. The outcome model $m_0(x)$ and the propensity score $e_0(x)$ are estimated by CatBoost¹ gradient boosting with $K = 5$ -fold cross-fitting. Propensity estimates are clipped to $[10^{-3}, 1 - 10^{-3}]$ for numerical stability.

Step 2: Residualisation. Out-of-fold residuals $\tilde{Y}_i = Y_i - \hat{m}^{(-k(i))}(X_i)$ and $\tilde{A}_i = A_i - \hat{e}^{(-k(i))}(X_i)$ are computed.

Step 3: Second-stage regression. Following the R-learner rearrangement [4], we define pseudo-outcomes $Z_i = \tilde{Y}_i / \tilde{A}_i$ and weights $w_i = \tilde{A}_i^2$, discarding observations with $|\tilde{A}_i| < 0.01$, and fit a weighted CatBoost regression of Z_i on X_i . Regularisation is implicit in the limited tree depth, shrinkage, and ordered boosting.

Remark 6.1. As noted in Section 7, L^∞ -convergence rates for gradient boosting are not established in general. The experiments below therefore constitute an empirical validation of $O_P(r_n + h^\alpha)$.

6.2.2 Bootstrap scale estimation

To estimate the noise level $\|\hat{\tau}_n - \tau_0\|_\infty$ empirically, we perform $B = 20$ bootstrap replications: resample with replacement, re-estimate $\hat{\tau}_n^{(b)}$, and evaluate at mesh vertices. From the $(B \times |\text{vertices}|)$ prediction matrix we extract:

- **Bootstrap amplitude:** mean of $Q_{0.99} - Q_{0.01}$ across replicates (robust range of $\hat{\tau}_n$).
- **Sup-discrepancy $\hat{\varepsilon}_n$:** the median of $\max_v |\hat{\tau}_n^{(b)}(v) - \bar{\tau}(v)|$, serving as an empirical proxy for r_n in the signal-to-noise criterion of Section ??.

6.2.3 Triangulation and interpolation

A fixed Delaunay triangulation $\mathcal{T}_h$ of $[0, 1]^2$ is constructed on a regular grid, producing a shape-regular mesh. Both the true function τ_0 and the estimator $\hat{\tau}_n$ are evaluated at the vertices of $\mathcal{T}_h$ and extended to piecewise-linear functions $\tau_{0,h}$ and $\hat{\tau}_{n,h}$ by barycentric interpolation.

6.2.4 Persistent homology and landscapes

Persistent homology in degree $q = 0$ is computed on the piecewise-linear functions $-\tau_{0,h}$ and $-\hat{\tau}_{n,h}$ via the sublevel-set filtration. This is equivalent to the superlevel-set filtration of $\tau_{0,h}$ and $\hat{\tau}_{n,h}$.

Persistence landscapes are computed on a common grid of 512 points covering the union of all birth-death intervals, with padding 0.1 on each side. The landscape distances are then computed numerically:

$$\|\Lambda_k(-\hat{\tau}_{n,h}) - \Lambda_k(-\tau_0)\|_{L^2} \approx \left(\sum_{j=1}^{512} |\lambda_k^{\hat{\tau}}(t_j) - \lambda_k^{\tau_0}(t_j)|^2 \cdot \Delta t \right)^{1/2}$$

¹CatBoost: <https://catboost.ai>. Hyperparameters: 500 iterations, depth 6, learning rate 0.05.

6.3 Experiment design

For each of the five topological scenarios, we run the full pipeline at four sample sizes:

$$n \in \{2000, 4000, 6000, 8000\}.$$

The triangulation mesh is held fixed across all sample sizes within each scenario, so that the geometric error $C_1 h^\alpha$ is constant and the observed decrease in the landscape distance is attributable solely to the decrease in the statistical error r_n .

For each $(n, \text{topology})$ pair, we record:

- the piecewise-linear nodal values f_{true} and f_{hat} at mesh vertices
- the persistence diagrams (PD) $\text{Dgm}_0(-\tau_{0,h})$ and $\text{Dgm}_0(-\hat{\tau}_{n,h})$, both in superlevel (f -coordinates) and sublevel ($-f$ -coordinates) representations
- the persistence landscapes (PL) $\Lambda_k(-\tau_{0,h})$ and $\Lambda_k(-\hat{\tau}_{n,h})$ on a common grid
- the landscape distance $\|\Lambda_k(-\hat{\tau}_{n,h}) - \Lambda_k(-\tau_{0,h})\|_{L^2}$
- bootstrap-based scale estimates for $\hat{\tau}_n$

6.4 Results

For each scenario we present:

- (a) function heatmap, persistence diagram and persistence landscape of the true function τ_0
- (b) persistence landscapes of $\hat{\tau}_{n,h}$ for each $n \in \{2000, 4000, 6000, 8000\}$.

Additionally, each experiment is repeated over 10 bootstrap replications, and the L^2 landscape distances are reported as mean $\pm$ standard deviation across replications.

6.4.1 Illustrations

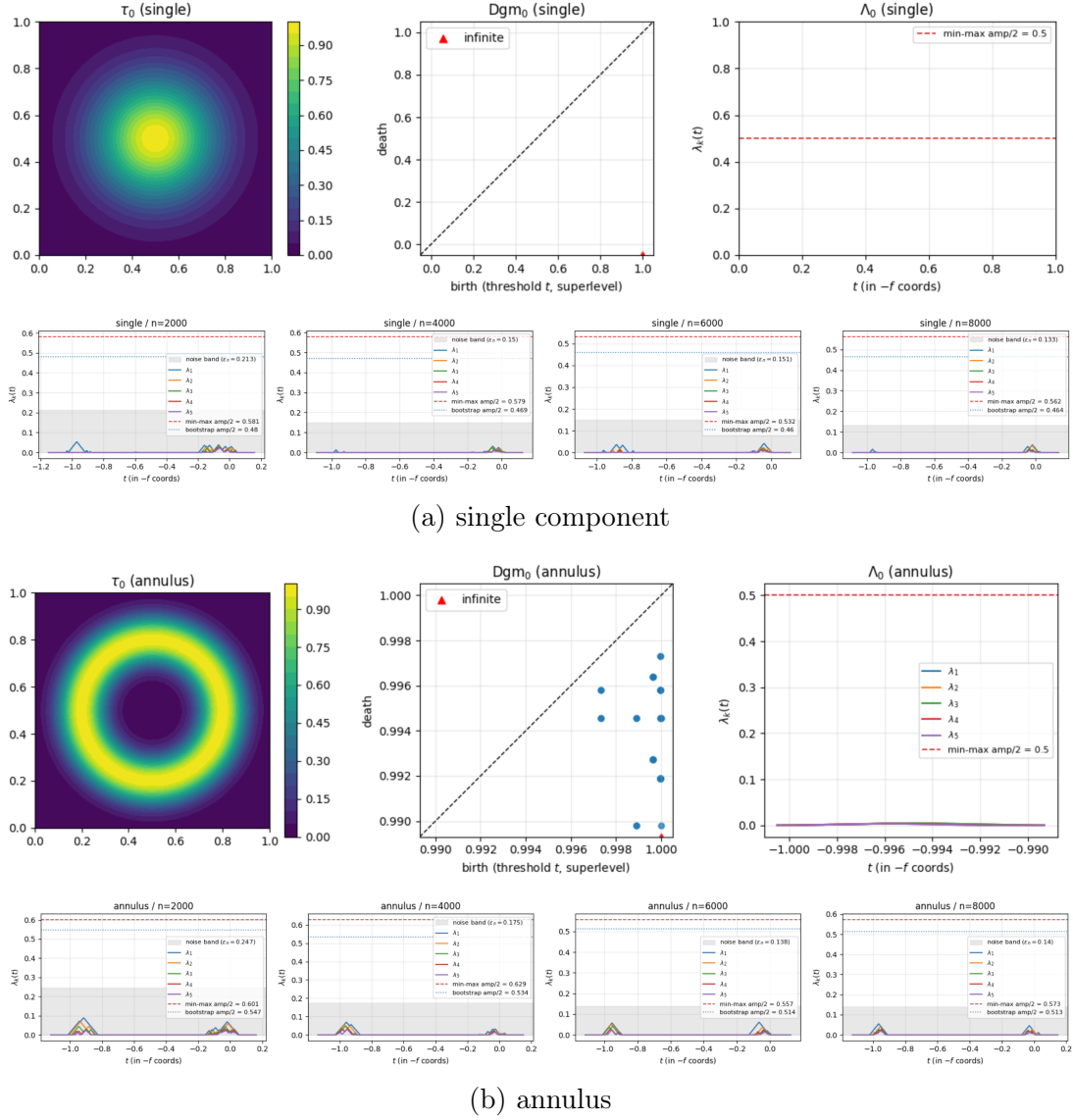
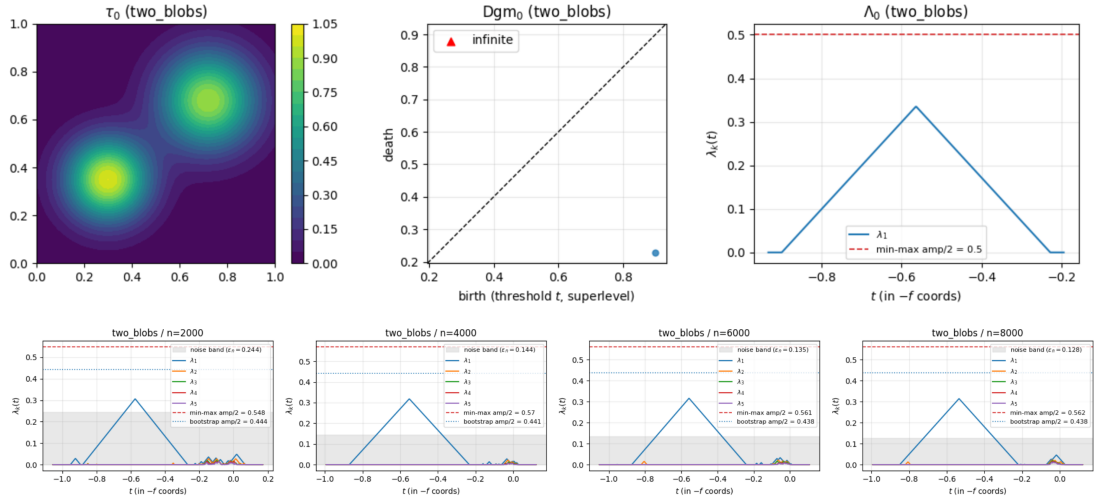


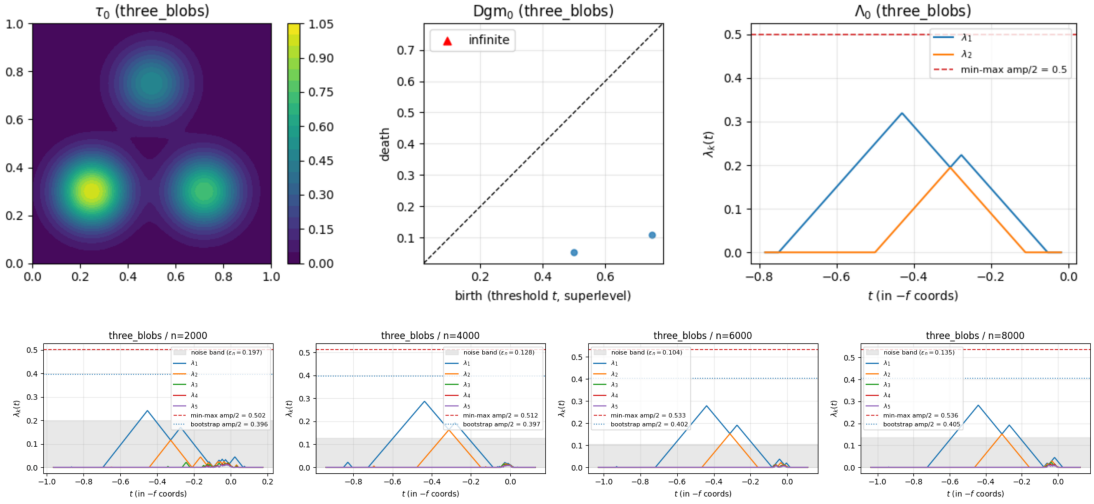
Figure 1: Empirical results for the single-component and annulus scenarios.

The single-component and annulus scenarios represent the simplest topological structures in our experiments, and the estimated persistence landscapes converge to the true ones rapidly.

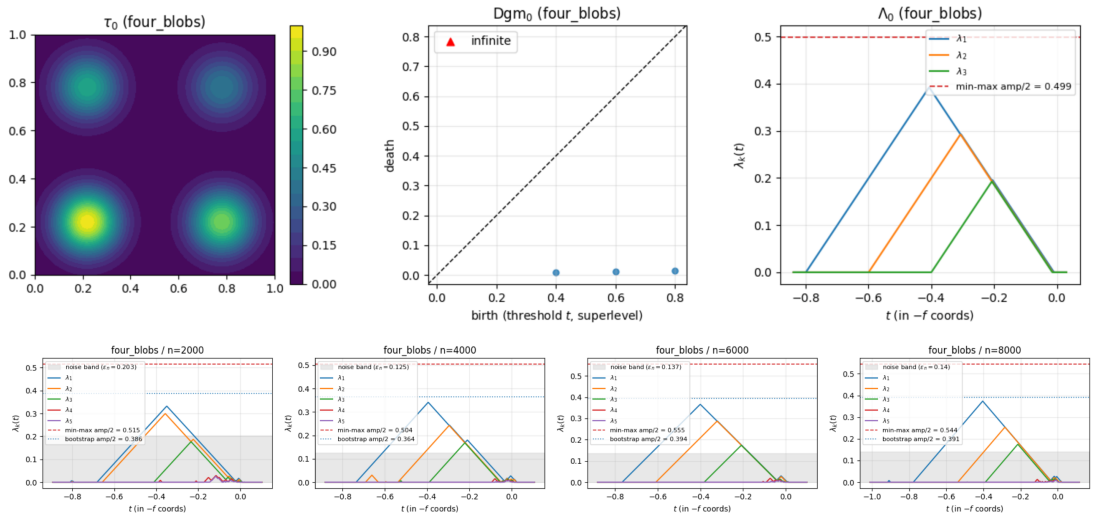
Already at $n = 2000$, the estimated persistence landscape $\Lambda_1(-\hat{\tau}_{n,h})$ closely matches the true landscape $\Lambda_1(-\tau_{0,h})$: the only features visible in the estimated diagram besides the dominant class are small-persistence points clustered near the diagonal, attributable to statistical noise. As n increases, these noise features only shrink further. This confirms that for simple topologies, a moderate sample size suffices to recover the correct topological structure.



(a) two connected components



(b) three connected components



(c) four connected components

Figure 2: Empirical results for the two-, three-, and four-component scenarios.

6.5 Convergence analysis

Table 2: Persistence landscape L^2 -distances for each scenario (mean $\pm$ std over 10 replications)

n	Single	Two	Three	Four	Annulus
2000	.020 $\pm$.007	.035 $\pm$.016	.068 $\pm$.012	.069 $\pm$.012	.043 $\pm$.009
4000	.013 $\pm$.007	.029 $\pm$.012	.049 $\pm$.011	.050 $\pm$.013	.028 $\pm$.003
6000	.012 $\pm$.004	.024 $\pm$.010	.044 $\pm$.012	.045 $\pm$.007	.024 $\pm$.005
8000	.011 $\pm$.009	.024 $\pm$.016	.043 $\pm$.006	.043 $\pm$.005	.020 $\pm$.004

As the sample size increases, the estimated persistence landscapes progressively converge to the true ones, although convergence is not immediate for non-trivial topologies as we can see in (Figure 2).

In all scenarios, the number of prominent peaks in the estimated persistence landscape matches the true number of connected components. At $n = 8000$, the positions, heights, and widths of the peaks align closely, the topological structure of τ_0 is faithfully recovered.

Table 2 shows that the average landscape distances decrease monotonically with n across all scenarios.

The standard deviations across bootstrap replications also decrease with n , the estimator becomes not only more accurate but also more stable with increasing sample size.

All of these observations are consistent with Theorem 5.12: since the triangulation is fixed (constant h), the observed decrease in landscape distances is driven by the reduction of the statistical error $r_n \rightarrow 0$ as $n \rightarrow \infty$. The monotone decrease, the variance reduction, and the eventual convergence to the true landscape are precisely the empirical manifestations of the bound $O_P(r_n + h^\alpha)$.

All empirical results are consistent with the theoretical predictions.

6.5.1 Limitations

- The experiments use a fixed triangulation; a systematic study of the geometric error (h -dependence) with varying mesh resolution would complement the current n -dependence analysis.
- CatBoost is used as the CATE estimator. While it performs well empirically, the theoretical rate r_n for gradient boosting in L^∞ -norm is not established (see the discussion in Section 7). The observed convergence is therefore an empirical validation rather than a proof of the rate.
- The sample sizes $n \in \{2000, \dots, 8000\}$ are moderate. Larger-scale experiments would allow a more precise estimation of the convergence exponent.
- All experiments are in $d = 2$. The curse of dimensionality implies slower rates for $d > 2$, which is not tested here.

7 Discussion

7.1 Optimality of the rate h^α

The exponent α in the geometric error bound is optimal:

- The interpolation error for $C^{0,\alpha}$ -functions is $\Theta(h^\alpha)$, not improvable without additional regularity.
- The bottleneck stability theorem is 1-Lipschitz (factor 1 before $\|f - g\|_\infty$).
- The L^∞ -stability of landscapes is 1-Lipschitz.
- The passage $L^\infty \rightarrow L^p$ does not change the power of h , only adds a constant.

Each step in the chain gives an unimprovable bound.

7.2 Optimal choice of mesh size

The total error $O_P(r_n + h^\alpha)$ is minimized when the two terms balance:

$$h^\alpha \asymp r_n \quad \implies \quad h \asymp r_n^{1/\alpha}.$$

For example, with $\tau_0 \in C^{0,\alpha}$, a kernel estimator, and the minimax rate $r_n = n^{-\alpha/(2\alpha+d)}$:

$$h_{\text{opt}} \asymp n^{-1/(2\alpha+d)}, \quad O_P(r_n + h^\alpha) = O_P(n^{-\alpha/(2\alpha+d)}).$$

7.3 Summation over landscape layers k

All estimates are given for a fixed layer k . For the full landscape norm $\|\Lambda\|_{L^p(\mathbb{N} \times \mathbb{R})}$, one needs to control the number of nonzero layers, which equals the number of off-diagonal points in the persistence diagram. For $-\hat{\tau}_{n,h}$ (PL on $\mathcal{T}_h$), this number can grow as $O(h^{-d})$.

The stability estimate (7) guarantees that “new” points (absent from $\text{Dgm}(-\tau_0)$) have persistence $\leq 2L h^\alpha$, i.e., they are noise. However, their *count* is not controlled by the current assumptions.

For fixed k , this issue does not arise.

7.4 On the realism of Hölder regularity

The assumption $\tau_0 \in C^{0,\alpha}(\Omega)$ means continuous dependence of the treatment effect on covariates. This is a standard condition in nonparametric statistics, satisfied whenever the causal mechanism is governed by continuous processes. The main situations where it fails are:

- Discrete covariates (excluded by $\Omega \subset \mathbb{R}^d$).
- Regression discontinuity designs (a different causal model, not considered here).
- Pathological oscillations (unrealistic for treatment effects).

7.5 On the strength of L^∞ -consistency

Assumption 7 requires uniform convergence of $\hat{\tau}_n$ to τ_0 in sup-norm. This is proven for:

Method	L^∞ -rate proven?
Kernel estimators	Yes
Spline/series estimators	Yes
Gradient boosting (XGBoost, LightGBM)	Not in general
Random forests	Not in general
Neural networks	Not in general

For methods without proven L^∞ -rates, one may consider the weaker condition

$$\max_{v \in \text{vertices}(\mathcal{T}_h)} |\hat{\tau}_n(v) - \tau_0(v)| = O_P(\tilde{r}_n)$$

which suffices for the interpolated estimator $\hat{\tau}_{n,h}$ and is closer to pointwise consistency.

7.6 Comparison with threshold-based clustering

A natural alternative to persistent homology is to fix a threshold t , construct the superlevel set $\{\hat{\tau}_n \geq t\}$, and apply standard clustering.

Property	Clustering	Persistence landscapes
Requires threshold choice	Yes	No
Stability guarantee	No	Yes (Theorem 5.12)
Lives in a linear space	No	Yes
Supports averaging	No	Yes
Supports hypothesis testing	Difficult	Yes (CLT, [7])
Detects higher topology (H_1 , holes)	No	Yes

The key advantage of persistent homology is that it considers *all thresholds simultaneously*, providing a threshold-free, stability-guaranteed summary of treatment heterogeneity. Theorem 5.12 makes this advantage quantitative.

8 Conclusion

The thesis has established a mathematical framework connecting causal inference with topological data analysis through the study of persistence landscapes of treatment effect functions.

The main result (Theorem 5.12) shows that the persistence landscapes of a triangulated statistical estimator of the treatment effect function converge to those of the true function at rate $O_P(r_n + h^\alpha)$, where r_n is the statistical estimation rate and h^α is the geometric approximation rate.

The framework provides a principled way to identify and validate *optimal treatment zones*—connected regions in the covariate space where treatment effects are pronounced and topologically stable.

The implementation is publicly available at:

github.com/ableenk/bachelor-thesis

8.1 Directions for future work

- **Higher homological degrees.** The current framework applies to arbitrary degree q , but the analysis and interpretation focus on $q = 0$. The case $q = 1$ (detecting “holes” in treatment effect regions) is of independent interest.
- **High-dimensional covariates.** Triangulation is computationally feasible only for $d \leq 4$. Extensions using alternative complexes (Vietoris–Rips, Čech) or manifold learning techniques would broaden the applicability.
- **Confidence bands for landscapes.** Combining Theorem 5.12 with bootstrap or multiplier bootstrap methods to construct confidence bands for persistence landscapes.
- **Weakening Assumption 7.** Replacing uniform consistency in L^∞ -norm by point-wise consistency at vertices of the triangulation, which is applicable to a wider class of estimators.

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